

The Spence Loss: Well-Conditioned Value-Function Learning for Reward-Guided Diffusion

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Abstract

Reward-guided diffusion and diffusion fine-tuning are, through entropy-regularized stochastic control, the problem of learning a value function $V(x, t) = \log \mathbb{E}_{\text{base}}[\exp r(X_1) \mid X_t = x]$ —a conditional log-partition function whose gradient is the optimal control. The unbiased Monte-Carlo target for V lives in exponential space, so squared-error regression has gradients that explode for over-predictions and, worse, vanish for under-predictions, where the model is most wrong and gradient clipping cannot help. We observe that every Bregman divergence has the value function as its population minimizer, and that such a loss reaches the optimizer only through a scalar *gradient weight* $w(u) = u\varphi''(u)$ on the exponential-space residual—so the design space is exactly the choice of w , with squared error ($w = 2u$) and the Itakura–Saito divergence ($w = 1/u$) as its two extremes. We then solve for the weight $w(u) = \ln u/(u-1)$ whose gradient is linear in the log-space prediction in *both* tails, yielding the *Spence loss*¹: a closed-form, numerically stable gradient under which neither over- nor under-predictions blow up or die. A secondary contribution analyzes why high-dimensional value learning is signal-sparse and studies on-policy importance-sampling schemes. We validate on 2-moons, a dimensional-scaling benchmark designed for clean statistical reads, and GMM-40, where our log-partition estimate is competitive with published diffusion samplers.

1 Introduction

2 value functions for reward-guided diffusion

3 The problem with exponential-space squared error

The value $V(x, t) = \log \mathbb{E}_{\text{base}}[e^{r(X_1)} \mid X_t = x]$ is a conditional log-partition function, and the H-martingale property (§2) supplies *conditionally unbiased* targets in *exponential* space: $e^{V(x_t)} = \mathbb{E}[e^q \mid x_t]$ for a one-step target q (the terminal reward, or a bootstrapped value). A network predicts the value in log space, $p = V_\theta(x_t)$; matching e^p to the unbiased target e^q with squared error gives

$$L_{\text{MSE}}(p, q) = (e^q - e^p)^2, \quad \frac{\partial L_{\text{MSE}}}{\partial p} = 2e^p(e^p - e^q) = 2e^{2p} - 2e^{p+q}. \quad (1)$$

Both tails are pathological, but asymmetrically so (Fig. 1, left).

Over-prediction explodes (recoverable). As $p \rightarrow +\infty$ the gradient grows like $2e^{2p}$ and overflows single precision once $p \gtrsim 44$ nats (the square needs e^{2p}). This is a nuisance but a benign one: the gradient merely needs to be a full-size, correctly-signed step, so saturating it to a large finite value (or clipping) recovers a usable update.

Under-prediction vanishes (unrecoverable). As $p \rightarrow -\infty$ the gradient behaves like $-2e^{p+q} \rightarrow 0$. This is exactly the regime where the model is *most wrong*: it assigns near-zero value $e^p \approx 0$ to a state whose true value e^q is large — a rare, high-reward target it has failed to find. The learning signal there is not merely ill-scaled, it is *absent*, and no amount of gradient clipping can manufacture a gradient that has decayed to zero. In the high-dimensional, signal-sparse regime of §6, where the useful targets are precisely the rare large ones, this is the dominant failure mode.

¹Named for the Spence (dilogarithm) function Li_2 appearing in its value; in our code and experiments it is the **quad** loss.

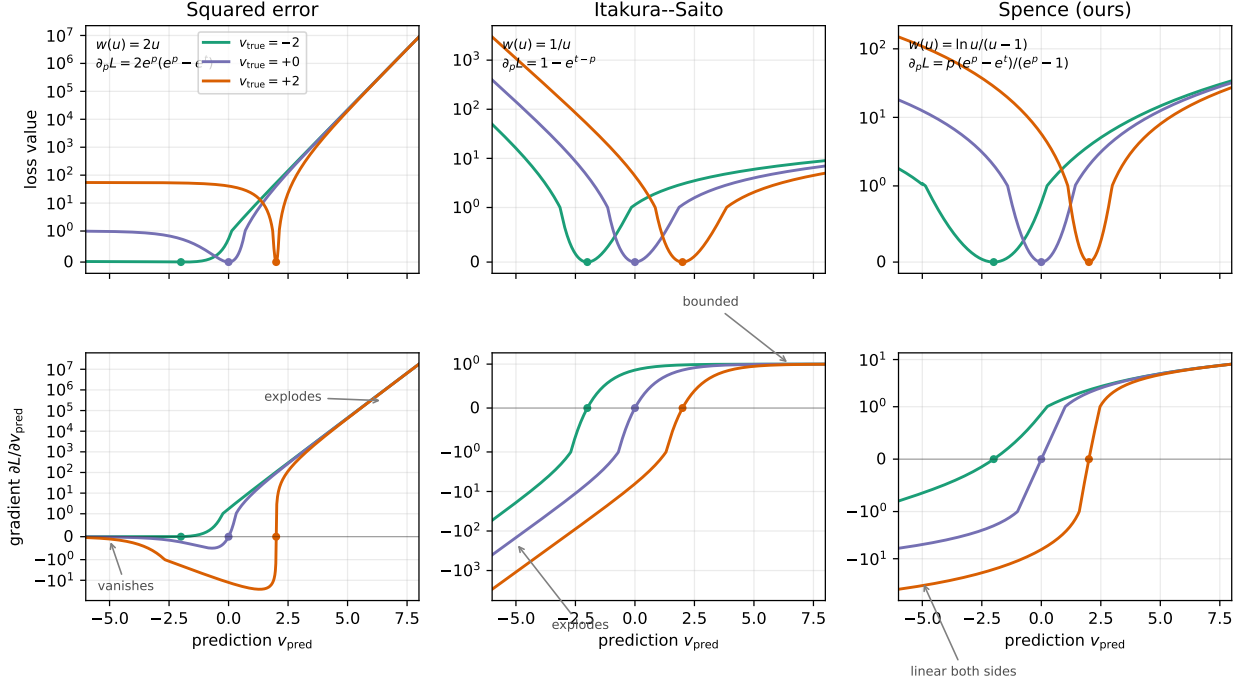


Figure 1: Loss value (top) and gradient $\partial L / \partial v_{\text{pred}}$ (bottom) of $D_F(e^{v_{\text{true}}}, e^{v_{\text{pred}}})$ as functions of the prediction v_{pred} , for three targets $v_{\text{true}} \in \{-2, 0, 2\}$ (dots mark the minimizer $v_{\text{pred}} = v_{\text{true}}$; symmetric-log axes). **Squared error** vanishes for under-predictions ($v_{\text{pred}} < v_{\text{true}}$, gradient $\rightarrow 0$) and explodes for over-predictions. **Itakura-Saito** inverts this: the over-prediction gradient saturates at +1, while under-predictions now explode — trading the unrecoverable pathology for a clippable one. **Spence (ours)** is linear in v_{pred} in both tails, so neither side vanishes nor explodes exponentially.

Why not regress in log space? The obvious escape is to drop the exponentials and fit $(p - q)^2$ directly, which is perfectly conditioned. It is also biased. Squared error in q is minimized by $\mathbb{E}[q \mid x_t]$, whereas the value is $V = \log \mathbb{E}[e^q \mid x_t]$; by Jensen the two differ by the log-partition gap

$$V(x_t) - \mathbb{E}[q \mid x_t] = \log \mathbb{E}[e^{q - \mathbb{E}[q]}] \geq 0, \quad \approx \frac{1}{2} \text{Var}(q \mid x_t) \text{ for small spread}, \quad (2)$$

and this gap is *largest* in exactly the heavy-tailed, high-dimensional regime the method is meant to handle: it is the quantity that a soft-max is all about, so discarding it discards the signal. Unbiasedness pins us to the exponential-space target; only the divergence applied to it is negotiable. That is the opening the Bregman family exploits (§4).

Empirical pathology. Beyond the asymptotics, the finite-precision behavior is severe. In our convergence runs the naive loss skipped or aborted a large fraction of batches on overflow — in one FBRRT run, 18% of batches once predictions reached moderate magnitude — and a *single* non-finite prediction propagated through the shared value network to poison every subsequent batch. We report the full stability audit in §5. A stabilized implementation with the exact gradient of (1) saturated to the float range (our `exp_mse`) removes the overflow and the poisoning, and it is the baseline we compare against throughout, so that the comparison isolates the divergence rather than the arithmetic. What it cannot remove is the vanishing gradient, which is a property of the divergence itself.

4 Bregman divergences for value learning

4.1 Any Bregman divergence learns the value function

For a strictly convex, differentiable potential φ on $(0, \infty)$, the Bregman divergence of a prediction u from a target T is

$$D_\varphi(T, u) = \varphi(T) - \varphi(u) - \varphi'(u)(T - u). \quad (3)$$

Squared error is the special case $\varphi(u) = u^2$, giving $D_\varphi(T, u) = (T - u)^2$. The order of the arguments matters: the prediction must occupy the *second* slot, which is the one whose minimizer is a plain mean.²

Lemma 1 (Bregman divergences are proper losses for the value). *Let the regression target be e^q , with q such that $\mathbb{E}[e^q | x_t] = e^{V(x_t)}$ (the H -martingale target, §2). Parameterize the prediction in log space, $u = e^p$. Then for any strictly convex φ ,*

$$\frac{\partial}{\partial p} D_\varphi(e^q, e^p) = w(e^p)(e^p - e^q), \quad w(u) := u \varphi''(u) > 0, \quad (4)$$

and consequently $\arg \min_p \mathbb{E}_{q|x_t}[D_\varphi(e^q, e^p)] = V(x_t)$.

Proof sketch. Differentiating (3) in u , the $\varphi'(u)$ terms cancel and leave $\varphi''(u)(u - T)$; the chain rule contributes $du/dp = u$. Because (4) is *linear* in the target, taking expectations gives $\mathbb{E}[\partial L / \partial p] = w(e^p)(e^p - \mathbb{E}[e^q | x_t])$, which is negative below e^V and positive above it. Full proof in App. D. $\square$

Two consequences drive the rest of this section. First, the choice of φ does *not* bias the learned value, so it is free to be chosen for *conditioning*. Second, (4) says that a Bregman loss acts on the optimizer only through the scalar *gradient weight* w multiplying the raw exp-space residual $e^p - e^q$. **The design space is exactly the choice of a positive weight function w on $(0, \infty)$** , and the two losses we have met are its opposite extremes.

4.2 Two reference points

Squared error: $w(u) = 2u$. The weight *grows* with the prediction, so $\partial L / \partial p = 2e^p(e^p - e^q)$ explodes like e^{2p} for over-predictions and, because $w(u) \rightarrow 0$ as $u \rightarrow 0$, is driven to zero for under-predictions — the unrecoverable pathology of §3.

Itakura–Saito: $w(u) = 1/u$. Taking $\varphi(u) = -\ln u$ gives $D_{\text{IS}}(e^q, e^p) = e^{q-p} - (q - p) - 1$ and

$$\frac{\partial L_{\text{IS}}}{\partial p} = 1 - e^{q-p}. \quad (5)$$

The weight decays fast enough to cancel the residual entirely on the over-prediction side: as $p \rightarrow +\infty$ the gradient saturates at $+1$, a bounded, correctly-signed step. On the under-prediction side it now *explodes* like $-e^{q-p}$. Itakura–Saito therefore does not remove squared error’s asymmetry so much as *invert* it, and that is precisely why it helps: it converts the unrecoverable failure (a gradient that has decayed to zero) into the recoverable one (a gradient that is too large, which clipping fixes). It is moreover *scale-invariant* ($D_{\text{IS}}(cT, cu) = D_{\text{IS}}(T, u)$, equivalently (5) depends on $q - p$ alone) and is the negative log-likelihood of a gamma/exponential observation model, giving a natural uncertainty interpretation for V . A useful midpoint, but still one-sided (Fig. 1, middle).

²With the arguments swapped, $\arg \min_u \mathbb{E}[D_\varphi(u, T)]$ solves $\varphi'(u) = \mathbb{E}[\varphi'(T)]$, a quasi-arithmetic mean — the harmonic mean of T for $\varphi = -\ln u$, not $\mathbb{E}[T]$.

4.3 The Spence loss

Rather than pick a potential and inspect its tails, we ask directly what weight w we want. Writing $u = e^p$ and reading off (4), the two requirements are:

$$p \rightarrow +\infty \ (u \rightarrow \infty) : \quad w(u) u \sim \ln u \quad \implies \quad w(u) \sim \ln u / u, \quad (6)$$

$$p \rightarrow -\infty \ (u \rightarrow 0) : \quad -w(u) e^q \sim e^q \ln u \quad \implies \quad w(u) \sim -\ln u, \quad (7)$$

i.e. in both directions the gradient should grow *linearly in the log-space prediction* $p = \ln u$. The simplest smooth positive function meeting both is

$$\boxed{w(u) = \frac{\ln u}{u - 1}} \quad \iff \quad \varphi''(u) = \frac{\ln u}{u(u - 1)}, \quad (8)$$

which is positive on all of $(0, \infty)$ (both factors change sign at $u = 1$) and extends smoothly through the removable singularity with $w(1) = 1$. It is the reciprocal of the logarithmic mean of 1 and u , and at both extremes it lies between the squared-error and Itakura–Saito weights: the Spence loss interpolates the two reference points rather than adding a third unrelated one. The choice is not unique — (6)–(7) fix only the asymptotics — but (8) is the member with a closed-form gradient and a dilogarithm value (App. A). Substituting it into (4) gives

$$\boxed{\frac{\partial L}{\partial p} = p \frac{e^p - e^q}{e^p - 1}} \quad (9)$$

zero iff $p = q$ (the apparent zero at $p = 0$ is cancelled by the pole of $p/(e^p - 1) \rightarrow 1$), with population minimizer V by Lemma 1.

Both tails linear in the prediction. By construction,

$$p \rightarrow +\infty : \quad \frac{\partial L}{\partial p} \rightarrow p, \quad \text{no } e^{2p} \text{ blow-up,} \quad (10)$$

$$p \rightarrow -\infty : \quad \frac{\partial L}{\partial p} \rightarrow p e^q, \quad \text{non-vanishing corrective signal.} \quad (11)$$

We stress what is and is not being claimed. The gradient is linear in the *prediction* p , not in the error $p - q$, so the loss grows like $\frac{1}{2}p^2$ for over-predictions and like $\frac{1}{2}p^2 e^q$ for under-predictions: it is quadratic in p on both sides, but with target-dependent curvature on the under-prediction branch. That e^q factor is the reason the loss is not scale-invariant, and it is deliberate — it up-weights exactly the rare high-value targets that dominate the log-sum-exp and are hardest to find (§6), while still growing only polynomially in p . To our knowledge this is the first Bregman divergence proposed for exp-space value regression with both tails controlled in this sense; the alternative strand of work attacks the same conditioning problem by changing the *objective* rather than the divergence (log-variance and adjoint losses, §8).

Value via the Spence function. The divergence value (used only for monitoring; training uses (9)) is expressible through the Spence function $S(x) = \text{Li}_2(1 - e^x)$, using the dilogarithm identity $S'(x) = -xe^x/(e^x - 1) = x/\text{expm1}(-x)$; hence the name. The full derivation and the numerically stable branch-wise gradient — finite for all float inputs, unlike naive autograd, which returns NaN for $p \gtrsim 88$ in single precision — are in App. A.

Implied observation model. By the duality between regular Bregman divergences and regular exponential families [1], minimizing any D_φ is maximum likelihood under the exponential family whose cumulant is the convex conjugate φ^* ; squared error recovers a Gaussian and Itakura–Saito a gamma. For (8) the conjugate is not a standard named family, so we report the correspondence rather than a closed-form likelihood; the practical content of the weight is the tail behavior above.

Table 1: The value-learning loss family, organized by the gradient weight w of (4). Tails give $\partial L/\partial p$ as the prediction $p \rightarrow \pm\infty$ at fixed target q . All three share the same minimizer, $V = \log \mathbb{E}[e^q \mid x_t]$ (Lemma 1); they differ only in conditioning.

loss	weight $w(u)$	under-predict ($p \rightarrow -\infty$)	over-predict ($p \rightarrow +\infty$)	scale-inv.	noise model
squared error	$2u$	<i>vanishes</i> ($\rightarrow 0$)	explodes (e^{2p})	no	Gaussian
Itakura–Saito	$1/u$	explodes ($-e^{q-p}$)	bounded ($\rightarrow +1$)	yes	gamma
Spence (ours)	$\frac{\ln u}{u-1}$	linear ($p e^q$)	linear (p)	no	implicit

5 Off-policy experiments: the loss in isolation

6 Signal sparsity and on-policy learning

7 Experiments

8 Related work

9 Conclusion and limitations

References

- [1] Arindam Banerjee, Srujana Merugu, Inderjit S. Dhillon, and Joydeep Ghosh. Clustering with Bregman divergences. *Journal of Machine Learning Research*, 6:1705–1749, 2005.

A The Spence loss: derivation and implementation

B On-policy method zoo

C Experimental details

D Additional proofs